

# RRH-CRMS REQUIREMENTS

## FOR RRH + SONTHILLU WEBSITES

CRM → Website data, Website → CRM data, business rules and publication requirements

Version 1.0 • Discussion-derived working specification

14 August 2026

Purpose: define the CRM-side information and integration requirements for the two public websites. This document records confirmed decisions from the planning discussion. Where an exact CRM field, API route or enum was not discussed, it is marked as needing repository verification rather than guessed.

# 1. Document Control

| Item                     | Value                                                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| Purpose                  | Define what RRH-CRMS must provide to the RRH and Sonthillu websites and what the websites must send back. |
| Consumers                | RRH website and Sonthillu website                                                                         |
| CRM                      | RRH-CRMS                                                                                                  |
| Property source of truth | RRH-CRMS                                                                                                  |
| Customer accounts        | Separate RRH and Sonthillu customer databases/accounts                                                    |
| Website content language | English                                                                                                   |
| AI Search language       | Multilingual and mixed-language                                                                           |
| Public availability rule | Reserved and Sold inventory are not live public listings                                                  |
| Document status          | Working requirements; final field/API names must be reconciled with actual CRM                            |

## 1.1 Rules for interpreting this document

- CONFIRMED = explicitly decided in the discussion.
- PROPOSED/CONCEPTUAL = structure needed to implement confirmed behavior; exact implementation must match CRM.
- TBD = not specified yet; must be verified before development.
- No technical assumption should override an existing CRM business rule.

# 2. Business Scope Relevant to CRM

## 2.1 RRH

- Commercial properties, agricultural land, open plots and related real-estate categories.
- Rentals are not handled.
- Both company/own inventory and resale/owner-submitted properties are supported.
- Current operating geography is Telangana; the model must support future state expansion.

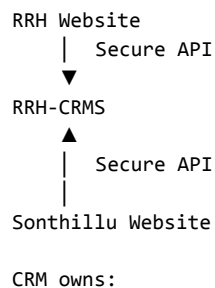
## 2.2 Sonthillu

- Residential/housing properties including apartments/flats, independent houses and villas.
- Rentals are not handled.
- Both company/own inventory and resale/owner-submitted properties are supported.
- Current operating geography is Telangana; the model must support future state expansion.

## 2.3 Connected brands

- RRH and Sonthillu are separate companies/brands with separate domains, branding and customer accounts.
- They are connected through the same CRM ecosystem.
- Cross-branding is approved by the owner.
- A property can be associated with both companies, and publication can be independent per brand.

# 3. System Boundary



- authoritative property/project data
- lifecycle and availability
- verification
- publication approval
- internal/source/seller information
- business workflows

Websites own:

- public presentation
- public search experience
- customer account experience
- AI Search interface
- recommendations
- public enquiries
- website analytics

Websites must not directly read/write the CRM database. Any website search index/cache is derived data and must not become a competing business source of truth.

## 4. CRM → Website Requirements

### 4.1 Live inventory eligibility

- The website API must return only properties that the current brand is authorized to publish.
- The property must have completed the CRM approval/publication requirements for that brand.
- The property must be currently Available to appear as live purchasable inventory.
- Reserved properties must not be returned as live inventory.
- Sold properties must not be returned as live inventory.
- Draft, rejected, review-only or otherwise unpublished records must not be exposed.
- Brand scoping must be enforced server-side, not only by frontend filtering.
- Seller identity and internal source type must not be part of public website payloads.

### 4.2 Independent brand publication

Property A

RRH publication = Published  
 Sonthillu publication = Not Published

Property B

RRH publication = Published  
 Sonthillu publication = Published

The CRM must be capable of this independent publication model because a shared physical property/project may be handled by both companies.

### 4.3 Public property information

| Data group          | Website requirement                                   | Internal-only?       |
|---------------------|-------------------------------------------------------|----------------------|
| Identity            | Stable property ID/reference, title, slug             | No, safe fields only |
| Classification      | Property type/category, New/Resale                    | No                   |
| Company association | Brand-specific public eligibility                     | Controlled           |
| Price               | Minimum and maximum/price range in INR                | No                   |
| Location            | State/city/locality + safe public location            | No                   |
| Exact location      | Exact address/coordinates                             | Yes                  |
| Size                | Built-up/carpet/plot area/acreage as applicable       | No                   |
| Specifications      | BHK, floor, facing, road width, etc. where applicable | Usually no           |
| Amenities/features  | Applicable public amenities                           | Usually no           |

|                         |                                          |                                 |
|-------------------------|------------------------------------------|---------------------------------|
| RERA                    | Public verified badge/status as approved | Public subset                   |
| RERA documents          | Detailed documents/checks                | Yes                             |
| Media                   | Approved photos/videos                   | Safe public media               |
| Availability            | Live inventory only when Available       | Internal states retained in CRM |
| Verification            | Safe public badge/indicator              | Detailed checks internal        |
| Source type             | Owner/company/channel-partner origin     | Yes                             |
| Seller details          | Seller identity/contact                  | Yes                             |
| Internal notes/approval | CRM-only notes/comments                  | Yes                             |

## 5. Property & Project Data Requirements

### 5.1 Property taxonomy

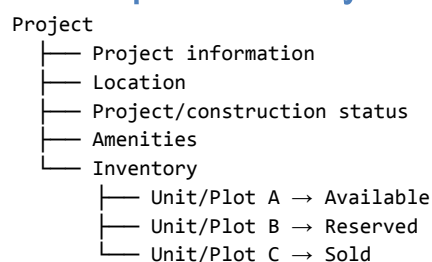
The exact canonical taxonomy must be read from the current CRM. The websites need stable IDs plus safe public labels.

| Brand    | Known categories from the discussion                                                                |
|----------|-----------------------------------------------------------------------------------------------------|
| RRH      | Commercial; Agricultural Land; Open Plots; other RRH categories; New and Resale variants            |
| Sonhillu | Apartments/Flats; Independent Houses; Villas; other residential categories; New and Resale variants |

### 5.2 Standalone + project inventory

- Support standalone properties.
- Support projects/developments.
- Apartment projects may contain towers/buildings, floors and units.
- Plot projects/layouts may contain individual plots.
- Project pages require project-level data plus current available inventory.
- The exact project/unit model must be reconciled with existing CRM tables/models.

### 5.3 Conceptual hierarchy



## 6. Status & Publication Requirements

### 6.1 Public availability

| Inventory state | CRM retains? | Live public listing?                 |
|-----------------|--------------|--------------------------------------|
| Available       | Yes          | Yes, if published/approved for brand |
| Reserved        | Yes          | No                                   |
| Sold            | Yes          | No                                   |

## 6.2 Separate concepts

Availability, project lifecycle/construction status and website publication should remain separate concepts. Exact enum names must be taken from the existing CRM.

```
Conceptual:
availabilityStatus = AVAILABLE | RESERVED | SOLD
projectStatus      = existing CRM project lifecycle
constructionStatus = existing CRM construction lifecycle
publicationStatus  = existing CRM publication/approval lifecycle
```

## 6.3 Publication prerequisites

- Required CRM verification complete.
- Required Project Manager/site verification complete for submitted resale/new property.
- Digital Marketing review complete where required.
- MD approval complete where required.
- Property marked published for the relevant brand.
- Property currently Available.

# 7. Seller / Resale Workflow Requirements

## 7.1 Website intake to CRM

```
Seller
↓
Website seller registration/intake
↓
Sell New Property or Resell Property
↓
Basic details
↓
CRM intake
↓
Project Manager assignment
```

The public seller form does not require images during initial submission.

## 7.2 CRM Project Manager stage

- Receive assigned seller submission.
- Visit the property.
- Verify property and seller-provided information.
- Update/correct property details where necessary.
- Add or update amenities.
- Capture property photographs.
- Capture exact property location in CRM.
- Check applicable RERA and other verification information.
- Complete internal verification notes/checks.

## 7.3 Digital Marketing stage

- Review property photographs.
- Check public text/spelling.
- Check presentation/marketing readiness.

## 7.4 MD approval stage

- Final management approval before publication.
- Only approved/published inventory becomes visible on the relevant website.

## 7.5 Seller vs verified data

The workflow requires CRM to distinguish initial seller input from the final verified/current property information wherever the existing CRM design permits. The PM may modify seller-provided values after verification.

# 8. CRM Location Capture Requirements

Location capture is an internal CRM operation during Project Manager site verification. It is not part of the public seller intake workflow.

- Provide a CRM 'Capture Current Location' action.
- Show clear instructions to stand at the actual property location before capture.
- Request device/browser location permission with an explanatory message.
- Store latitude.
- Store longitude.
- Store location accuracy when available.
- Store capture timestamp.
- Store the employee identity responsible for capture.
- Allow PM to manually verify/correct address/public location representation.
- Keep exact coordinates internal; the public website receives only a safe approximate location.

# 9. Price Requirements

- All listed real-estate prices are negotiable; customer-facing UI should not falsely imply fixed transaction rates.
- Property/project price may be represented as a range such as ₹45L – ₹53L.
- CRM should expose structured numeric price boundaries in INR to allow search and ranking.
- Minimum/maximum search budget is a customer requirement, not a promise of final transaction price.

Examples:

max budget ₹50L → max = 5,000,000  
₹40L–₹60L → min = 4,000,000; max = 6,000,000  
above ₹1Cr → min = 10,000,000

# 10. Location & Geographic Search Data

- Launch market is Telangana.
- Schema must support future additional states.
- CRM should expose structured state/city/locality/project/location identifiers where available.
- Public search may use safe location data for map and proximity functions.
- Exact property coordinates remain internal.
- For future 'near metro/landmark' matching, the website/search layer needs a reliable relationship or coordinate dataset. CRM should provide enough canonical property location data to support that when the geographic dataset is available.

Visakhapatnam was used only as a sample location in the discussion; it is not current launch scope.

## 11. Search Requirements from CRM

### 11.1 Normal Search

- Location.
- Property category/type.
- New/Resale/Any.
- Minimum budget.
- Maximum budget.
- Property-type-specific searchable attributes.
- Amenities/features where supported.
- Project/unit relationship.
- Availability/publication.
- Safe public location.
- Approved media.

### 11.2 AI Search data contract

The AI does not need to understand the CRM schema directly. It converts natural-language input into structured requirements. CRM must expose canonical values/relationships that the website matching engine can query.

| AI requirement               | CRM data needed                                          |
|------------------------------|----------------------------------------------------------|
| Budget                       | Structured price min/max in INR                          |
| Property type                | Canonical type IDs/labels                                |
| BHK                          | BHK field for applicable residential properties          |
| Location                     | Canonical state/city/locality/project identifiers        |
| Near landmark                | Coordinates/geo relationship or supported location index |
| Amenities                    | Canonical amenity IDs + property associations            |
| Area/size                    | Numeric value + unit                                     |
| New/Resale                   | Canonical classification                                 |
| Availability                 | Current availability state                               |
| Project                      | Project ↔ inventory relationship                         |
| Other property-specific need | Canonical field when business supports it                |

### 11.3 Applicability

CRM/search data must distinguish Not Applicable from Missing. For example, agricultural land has no BHK; that should not be represented as a failed/false BHK value.

## 12. Dynamic Match & Ranking Engine Data Requirements

The matching algorithm must be dynamic and requirement-aware. Fixed global percentages are not acceptable as the universal model because different users express different requirements and different property types have different fields.

### 12.1 Requirement classes

| State             | Meaning                                |
|-------------------|----------------------------------------|
| Hard              | User explicitly requires the condition |
| Strong preference | Important but can be compromised       |
| Soft preference   | Useful but not essential               |
| Flexible          | User explicitly permits variation      |
| Unknown           | User did not specify                   |
| Not applicable    | Field does not apply to property type  |
| Excluded          | User explicitly does not want it       |

### 12.2 Data quality required for matching

- Missing must remain distinguishable from False.

- Not Applicable must remain distinguishable from Missing.
- Verified/final values should be distinguishable from unverified seller-entered values where needed.
- Availability must be current.
- Price must be numeric and structured.
- Location must be structured enough for geographic matching.
- Applicable property-specific attributes must be queryable.

### 12.3 Example behavior

If a customer searches for a 2BHK, an agricultural plot should not receive a poor BHK score; BHK is not applicable and the property is outside the intended property-type candidate set. If a residential apartment has BHK missing, that is unknown and should be handled differently from a verified non-match.

If a customer sets a maximum budget of ₹50L, exact/below-budget properties rank first. Slightly-above-budget properties may be shown as related alternatives, clearly labeled as above budget, but must not silently replace exact matches.

## 13. Search Recommendation Requirements

### 13.1 Required V1 recommendation data

- Stable property IDs.
- Property type/category.
- Location.
- Price.
- Applicable specifications.
- Amenities/features.
- Availability.
- Publication state.
- Project relationship.
- Approved media.

### 13.2 Recommendation categories

- Exact/strong match results.
- Related properties.
- Slightly above-budget alternatives.
- Nearby-location alternatives.
- Alternative property types where valid.
- Recently viewed properties.
- Popular/featured properties for first-time visitors.
- No-result recovery suggestions.

### 13.3 Brand isolation

- RRH recommendations use RRH-approved live inventory.
- Sonthillu recommendations use Sonthillu-approved live inventory.
- A shared property is returned to a website only if it is separately published/eligible for that website.
- Reserved and Sold properties do not appear as live recommendations.

## 14. Website → CRM Enquiry Requirements

### 14.1 Request a Call

| User      | Website submits                                                                     | CRM outcome                    |
|-----------|-------------------------------------------------------------------------------------|--------------------------------|
| Logged-in | Known customer context + preferred contact time + property/project context          | Create/update CRM enquiry/lead |
| Guest     | Name/contact + preferred contact time + required consent + property/project context | Create/update CRM enquiry/lead |

### 14.2 Call Now

Both websites also provide a direct Call Now action. The exact phone/telephony configuration is a deployment/business setting, not a CRM property requirement.

### 14.3 Multi-property enquiry

- User may select multiple properties.
- Website sends selected property IDs plus customer/contact context.
- CRM should support a single enquiry/request referencing multiple selected properties if its existing model permits; otherwise an integration mapping must be defined.

### 14.4 Lead source

CRM should record whether the enquiry originated from RRH or Sonthillu website so management can compare website performance.

## 15. Separate Customer Accounts

- RRH and Sonthillu have separate customer login systems/databases.
- Website customer records are not silently merged across the two brands.
- Shortlist, compare and saved activity are scoped to the relevant brand account.
- Cross-brand navigation can send the user to the other website, where a separate account can be used/created.

The exact customer-to-CRM identity linking strategy must be verified against the existing CRM authentication/customer architecture.

## 16. Verification & RERA

- CRM stores detailed property verification and document information.
- RERA checking is part of internal verification where applicable.
- Website can show an approved public RERA Verified indicator/status.
- Detailed RERA documents and internal checks remain private to CRM.
- A missing document/field must not be interpreted as verified.

## 17. Data That Must Not Be Exposed Publicly

| CRM internal data           | Website response |
|-----------------------------|------------------|
| Seller identity/contact     | Exclude          |
| Source type                 | Exclude          |
| Project Manager identity    | Exclude          |
| Internal notes              | Exclude          |
| Detailed verification notes | Exclude          |
| Legal/internal documents    | Exclude          |
| Exact GPS coordinates       | Exclude          |
| Internal approval comments  | Exclude          |

|                                      |                                                     |
|--------------------------------------|-----------------------------------------------------|
| Internal channel-partner information | Exclude unless future requirement explicitly allows |
|--------------------------------------|-----------------------------------------------------|

## 18. Required CRM API Capabilities

Capability-level API requirements are below. Exact endpoint names and HTTP contracts must be taken from the existing CRM implementation.

| Capability                   | Direction                           | Requirement                                        |
|------------------------------|-------------------------------------|----------------------------------------------------|
| Live property search         | CRM → website                       | Brand-scoped, published + Available inventory only |
| Property detail              | CRM → website                       | Safe public details by ID/slug                     |
| Project search               | CRM → website                       | Published project records                          |
| Project detail               | CRM → website                       | Public project data + current available inventory  |
| Taxonomy/locations           | CRM → website                       | Canonical property types, locations and filters    |
| Amenities/attributes         | CRM → website                       | Applicable public search fields                    |
| Enquiry                      | Website → CRM                       | Create/update enquiry                              |
| Request a Call               | Website → CRM                       | Create/update callback request                     |
| Multi-property enquiry       | Website → CRM                       | Reference selected property IDs                    |
| Seller submission            | Website → CRM                       | Create initial seller/property request             |
| Publication/state update     | CRM → website                       | Propagate publication/availability changes         |
| Search/recommendation events | Website → analytics/CRM if required | Measure business-relevant behavior                 |

### 18.1 API quality requirements

- Versioning.
- Typed/schema-validated contracts.
- Authentication/authorization.
- Brand scope.
- Rate limiting.
- Safe retries for idempotent operations.
- Consistent errors.
- Correlation/request IDs.
- No internal fields in public DTOs.
- Monitoring and audit logging.

## 19. Publication & Availability Synchronization

```

CRM state
  ↓
Company-specific publication eligibility
  ↓
Website API / cache / event
  ↓
RRH or Sonthillu public inventory

Important transitions:
  published + available → public
  available → reserved → remove from live
  available/reserved → sold → remove from live
  unpublished/rejected → never expose

```

- The website must not rely on long-lived stale cache for availability.
- If a property changes to Reserved/Sold, public search must stop presenting it as available.
- If V1 lacks webhooks, use safe short-cache/revalidation behavior.

- CRM remains authoritative when website/cache and CRM disagree.

## 20. CRM Fields Required by Major Website Feature

| Website feature        | CRM data required                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------|
| Normal Search          | Location, type, New/Resale, budget, applicable attributes, amenities, availability, publication |
| AI Search              | Same canonical data plus searchable relationships and values                                    |
| Property card          | Title, primary photo, type, price range, location, key specs, availability, verification badge  |
| Property detail        | Full safe public details, media, amenities, RERA-safe data, approximate map                     |
| Project page           | Project data, media, status, amenities, available inventory, price range                        |
| Shortlist              | Stable property IDs + current availability check                                                |
| Compare                | Stable IDs + comparable public fields                                                           |
| Request a Call         | Property/project context + customer reference + preferred time                                  |
| Multi-property enquiry | Selected property IDs + customer context                                                        |
| Sell Property          | Seller intake endpoint + submission reference                                                   |
| Recommendations        | Stable IDs + structured searchable attributes + availability/publication                        |
| Analytics              | Stable IDs/type/location identifiers for aggregation                                            |

## 21. Edge Cases

| Case                                       | CRM requirement/behavior                                                    |
|--------------------------------------------|-----------------------------------------------------------------------------|
| BHK not applicable (e.g. land)             | Expose property type/applicability; do not treat BHK as false               |
| BHK missing on apartment                   | Return unknown/null; matching engine handles uncertainty                    |
| Seller value differs from PM value         | Public API returns current verified/final CRM value                         |
| Seller claims amenity, PM not verified     | Do not present as verified if verification is required                      |
| Published for RRH only                     | RRH can return; Sonthillu must not                                          |
| Published for both                         | Each website receives only its own eligible public representation           |
| Reserved/Sold                              | Not live                                                                    |
| Exact GPS exists                           | Keep private; return public-safe approximate location                       |
| Price missing                              | Return an explicit approved public state if supported; never invent a value |
| Property becomes unpublished after caching | Public detail/search should revalidate safely                               |
| CRM unavailable                            | Website fails safely and hides internal error details                       |
| Duplicate enquiry                          | Use idempotency/deduplication where practical                               |
| Cross-brand unauthorized record            | Never leak via search/recommendation                                        |

## 22. Observability & Audit

- Log website API health and failures in a privacy-safe manner.
- Track failed enquiry creation.
- Track failed seller submission intake.
- Track publication/state synchronization issues.
- Use correlation/request IDs across website → API → CRM.
- Audit sensitive publication/approval actions.
- Do not write seller PII or internal notes into general logs.

## 23. CRM Integration Security

- Public websites never connect directly to MySQL.
- Authenticated API boundary between websites and CRM.
- Server-side brand authorization.

- Least-privilege service credentials.
- IDOR protection.
- Rate limits.
- No CRM secrets in browser code.
- Separate public-read inventory endpoints from privileged internal write APIs.
- Sensitive seller/location data protected.

## 24. Must Verify in Existing RRH-CRMS Before Development

1. Actual property model/table and current lifecycle.
2. Project/tower/floor/unit/plot model.
3. Current availability and publication fields/enums.
4. Existing property APIs.
5. Existing project/inventory APIs.
6. Existing customer/lead APIs.
7. Existing site-visit workflow and APIs.
8. Project Manager assignment workflow.
9. Existing RERA/document fields.
10. Existing media storage and media APIs.
11. Existing GPS/location fields.
12. Existing authorization/service-account model.
13. Existing CRM host/API configuration.
14. Customer Portal integration boundary.
15. Whether independent RRH/Sonhillu publication already exists or must be added.

## 25. Explicit Non-Assumptions

- No fixed universal match percentages.
- No rentals.
- No seller-direct communication.
- No public exact GPS.
- No public seller/source information.
- No public internal legal documents.
- No final AI model chosen yet.
- No map provider chosen yet.
- No endpoint names are final.
- No exact CRM lifecycle enum names are final until inspected.
- No additional launch states beyond Telangana.
- Visakhapatnam is only an example, not current launch scope.

## 26. CRM Integration Acceptance Checklist

1. RRH website can retrieve only RRH-approved, published and Available inventory.
2. Sonthillu website can retrieve only Sonthillu-approved, published and Available inventory.
3. CRM can independently publish a shared property to either or both brands.
4. Reserved and Sold inventory cannot be exposed as live purchasable listings.
5. Public payload excludes seller identity and internal source type.
6. Exact coordinates are protected.
7. RERA public indicator is controlled by approved CRM data.
8. Price ranges are available in structured form.
9. New/Resale classification is available.
10. Applicable property-specific attributes are available.
11. Project and inventory relationships are available.
12. Normal Search can be built entirely from canonical CRM data.
13. AI Search can map normalized requirements onto canonical CRM data.
14. Recommendations can use brand-scoped live inventory.
15. Enquiry and Request-a-Call can be created in CRM.
16. Multi-property enquiry can reference multiple selected property IDs.
17. Seller submissions reach CRM without direct publication.
18. Project Manager verification remains a CRM workflow.
19. Digital Marketing review remains a CRM workflow.
20. MD approval remains a CRM workflow.
21. Publication/availability changes propagate to websites.
22. RRH/Sonthillu customer accounts remain separate.
23. Cross-brand data leakage is prevented.

## 27. Final Principle

RRH-CRMS is the business and property authority. The websites should ask CRM for approved public information and send customer actions back to CRM. They should not recreate the internal property lifecycle, seller verification, project-manager work, Digital Marketing review or MD approval in the websites.

For the websites to deliver Normal Search, AI Search, dynamic matching, recommendations, property details, project pages, shortlist/compare, enquiries and seller submission, CRM needs canonical, brand-scoped, structured and sufficiently complete property/project data.

Where the discussion did not establish an exact field or implementation, this document deliberately marks it for repository verification. The next technical step is to inspect the existing RRH-CRMS property schema, lifecycle and APIs and reconcile them against this requirement set.